

USER MANUAL

# Lab Snack Inventory Management System

## Non-Technical Guide for Admin, Manager, and Staff

|             |                                       |
|-------------|---------------------------------------|
| Document    | User Manual                           |
| Version     | 2.1                                   |
| Date        | June 2026                             |
| Application | Lab Snack Inventory Management System |
| Audience    | Admin, Manager, Staff                 |

Prepared as a user-friendly PDF manual for daily operation of the system.

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## 1. Introduction

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**Lab Snack Inventory Management System** is a web application for managing snacks in a university or research lab. It helps lab members and administrators keep snack stock organized, transparent, and easy to monitor.

The system helps users:

- Check available snacks and current stock.
- Take snacks while respecting daily limits for Staff users.
- Request more stock when items are running low.
- Approve or reject restock requests.
- Review snack activity history.
- View simple reports about stock, usage, and expiring items.
- Receive in-app alerts for low stock, pending requests, and expiring products.

The system is used through a web browser. Users do not need to install a mobile application.

This release does not include online payment, supplier ordering, email notifications, SMS notifications, or public self-registration. User accounts are created by an Admin.

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## 2. Getting Started

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### 2.1 What You Need

| Requirement | Details                                                     |
|-------------|-------------------------------------------------------------|
| Browser     | Chrome, Firefox, Edge, or Safari is recommended.            |
| Access link | Use the website link provided by your lab administrator.    |
| Account     | Use the email and password given by your lab administrator. |

| Requirement | Details                                                            |
|-------------|--------------------------------------------------------------------|
| Camera      | Optional. A camera is only needed when scanning a product QR code. |

## 2.2 Sign In

1. Open the website link in your browser.
2. If you are not signed in, the system will show the Login page.
3. Enter your email and password.
4. Select **Sign in**.
5. After a successful sign-in, the Dashboard will appear.

## 2.3 Sign Out

Select **Sign out** in the top-right area of the page. On smaller screens, open the menu first and then select **Sign out**.

Always sign out when using a shared computer.

## 2.4 Need an Account?

Self-registration is not available. Contact your lab Admin if you need a new account, a password reset, or a role change.

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# 3. User Roles

The system has three user roles: **Admin**, **Manager**, and **Staff**. Each role has different permissions.

### 3.1 Admin

Admin users have full access to the system. They can:

- View all pages.
- Add, edit, and delete products.
- Adjust stock quantities and low-stock thresholds.
- Import products from a spreadsheet file.
- Approve or reject restock requests.
- Manage user accounts and roles.
- Take snacks without Staff quota limits.

### 3.2 Manager

Manager users handle daily inventory operations. They can:

- View inventory, history, analytics, and notifications.
- Add and edit products.
- Adjust stock quantities and low-stock thresholds.
- Import products from a spreadsheet file.
- Approve or reject restock requests.
- Take snacks without Staff quota limits.

Managers cannot delete products and cannot manage user accounts.

### 3.3 Staff

Staff users can:

- View available snacks.

- Take snacks within the Staff quota.
- Scan product QR codes.
- Create restock requests.
- View their dashboard, history, analytics, and notifications.

Staff users cannot add, edit, or delete products. They also cannot approve restock requests or manage users.

### 3.4 Staff Snack Limits

| Limit                  | Default Value |
|------------------------|---------------|
| Maximum items per take | 3 items       |
| Maximum items per day  | 10 items      |

Admin and Manager users are not limited by these Staff snack limits.

### 3.5 Permission Summary

| Feature                            | Admin | Manager | Staff |
|------------------------------------|-------|---------|-------|
| Sign in                            | Yes   | Yes     | Yes   |
| View Dashboard                     | Yes   | Yes     | Yes   |
| View Inventory                     | Yes   | Yes     | Yes   |
| Take snacks                        | Yes   | Yes     | Yes   |
| Scan QR code                       | Yes   | Yes     | Yes   |
| Add or edit products               | Yes   | Yes     | No    |
| Delete products                    | Yes   | No      | No    |
| Adjust stock                       | Yes   | Yes     | No    |
| Import products                    | Yes   | Yes     | No    |
| Create restock requests            | Yes   | Yes     | Yes   |
| Approve or reject restock requests | Yes   | Yes     | No    |
| View History                       | Yes   | Yes     | Yes   |
| Filter History by user             | Yes   | Yes     | No    |
| View Analytics                     | Yes   | Yes     | Yes   |
| Manage users                       | Yes   | No      | No    |

## 4. Navigation

After signing in, the main page contains three important areas.

| Area    | What It Does                                                                                        |
|---------|-----------------------------------------------------------------------------------------------------|
| Header  | Shows the logo, search bar, theme button, notification bell, user information, and sign-out option. |
| Sidebar | Provides quick access to main pages on desktop screens.                                             |

| Area         | What It Does                          |
|--------------|---------------------------------------|
| Main content | Shows the current page you are using. |

### 4.1 Main Menu

| Menu      | Purpose                                                                    | Who Can Use It |
|-----------|----------------------------------------------------------------------------|----------------|
| Dashboard | View a role-based overview.                                                | All roles      |
| Inventory | View snacks, take snacks, scan QR codes, and manage products when allowed. | All roles      |
| History   | Review inventory activity.                                                 | All roles      |
| Requests  | Create or review restock requests.                                         | All roles      |
| Analytics | View reports and charts.                                                   | All roles      |
| Users     | Create, edit, or delete user accounts.                                     | Admin only     |

### 4.2 Search

Use the search bar to find snacks by name or SKU. Search is helpful when the inventory list is long.

### 4.3 Theme Button

Use the theme button in the header to switch between light and dark appearance.

## 5. Dashboard

The Dashboard is the first page shown after sign-in. It gives a quick overview based on your role.

### 5.1 Staff Dashboard

Staff users can see:

- **Remaining today** - how many snacks can still be taken today.
- **Used today** - how many snacks have already been taken today.
- **Max per take** - the maximum number of items allowed in one take action.
- **My requests** - recent restock requests submitted by the user.
- **Popular snacks** - commonly taken items.

### 5.2 Manager Dashboard

Manager users can see:

- Total number of products.
- Number of pending restock requests.
- Number of low-stock alerts.
- Recent restock activity.
- Requests waiting for approval.

### 5.3 Admin Dashboard

Admin users see the same information as Managers, plus the total number of user accounts.

## 6. Inventory

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The Inventory page shows all snack products as cards in a grid.

### 6.1 View Snacks

Each product card may show:

- Product image.
- Product name.
- Category.
- SKU.
- Current stock quantity.
- Unit, such as piece, bottle, pack, or box.
- A **LOW STOCK!** badge when the product is at or below its alert level.
- An expiry warning when the product is close to its expiry date.

### 6.2 Filter Snacks

Use the filter buttons at the top of the Inventory page.

| Filter    | What It Shows                                     |
|-----------|---------------------------------------------------|
| All       | All products.                                     |
| Low Stock | Products at or below their low-stock alert level. |
| Expiring  | Products that are close to expiry.                |

### 6.3 Take a Snack

1. Open the Inventory page.
2. Find the snack you want.
3. Use the plus or minus buttons to choose the quantity.
4. Select **Take snack**.
5. The product stock is updated immediately.
6. The action is recorded in History.

For Staff users, the system checks the per-take and daily limits. If the quantity is too high or there is not enough stock, the system will show a clear message and the action will not be completed.

Admin and Manager users can take snacks without Staff quota limits.

### 6.4 Scan a QR Code

If your device has a camera:

1. Select **Scan SKU** on the Inventory page.
2. Allow camera access when your browser asks for permission.
3. Point the camera at the product QR code.
4. The matching product will be found and highlighted.

### 6.5 Product Details

Select a product name or image to open the product detail page. The detail page shows more information about the product, including stock level and product-specific history.

Admin and Manager users can also edit product details, adjust quantity, change the low-stock alert level, and upload a product image.

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## 7. Product Management

Product management is available to Admin and Manager users. Staff users can view products but cannot create or edit them.

### 7.1 Add a New Snack

1. Open Inventory.
2. Select **Add Snack**.
3. Fill in the required information: product name, SKU, and category.
4. Add optional information such as description, image, unit, and expiry date.
5. Set the initial quantity.
6. Set the low-stock alert level.
7. Select **Create snack**.

The SKU must be unique. If another product already uses the same SKU, the system will ask you to choose a different one.

### 7.2 Edit a Product

1. Open the product detail page from Inventory.
2. Update the product information.
3. Save the changes.

Admin users can delete a product. Manager users cannot delete products.

### 7.3 Adjust Stock Manually

Admin and Manager users can adjust stock from the product detail page.

| Action                       | Meaning                                                |
|------------------------------|--------------------------------------------------------|
| Set quantity                 | Directly update the current stock number.              |
| Change low-stock alert level | Decide when the product should be marked as low stock. |

Manual changes are recorded in History so that the lab can review what changed and who changed it.

### 7.4 Import Products from a Spreadsheet

Admin and Manager users can add many products at once using the import feature.

1. Open Inventory.
2. Select **Import CSV**.
3. Download the template file.
4. Fill in the product information in the template.
5. Upload the completed file.
6. Review the import result.
7. Check the Inventory page to confirm that the products were added.

The template includes these fields:

| Field    | Required | Description                                        |
|----------|----------|----------------------------------------------------|
| SKU      | Yes      | Unique product code.                               |
| Name     | Yes      | Product name.                                      |
| Category | Yes      | Product category, such as Drink, Snack, or Bakery. |

| Field                 | Required | Description                                                             |
|-----------------------|----------|-------------------------------------------------------------------------|
| Quantity              | Yes      | Starting stock quantity.                                                |
| Low-stock alert level | Yes      | Stock number that triggers a low-stock alert.                           |
| Unit                  | No       | Product unit, such as piece, box, pack, bottle, cup, kilogram, or gram. |
| Unit label            | No       | Display label, such as bottle or pack.                                  |
| Expiry date           | No       | Product expiry date.                                                    |
| Description           | No       | Short product description.                                              |

If some rows fail, the system will show which rows need correction. Common issues include duplicate SKU, missing required fields, or invalid values.

## 8. Restock Requests

The Requests page is used to ask for more stock and to approve or reject requests.

### 8.1 Create a Restock Request

All roles can create a restock request.

1. Open Requests.
2. Choose the product that needs more stock.
3. Enter the requested quantity.
4. Select **Submit request**.
5. The request will be marked as **Pending**.

Admin and Manager users will receive an in-app notification when a new request is submitted.

### 8.2 Approve a Restock Request

Admin and Manager users can approve pending requests.

1. Open Requests.
2. Find a request marked **Pending**.
3. Select **Approve**.
4. Confirm the action.

After approval, the product stock increases automatically and the requester receives a notification.

### 8.3 Reject a Restock Request

Admin and Manager users can reject pending requests.

1. Open Requests.
2. Find a request marked **Pending**.
3. Select **Reject**.
4. Optionally enter a reason.
5. Confirm the rejection.

After rejection, the stock does not change and the requester receives a notification.



## 8.4 Request Statuses

| Status   | Meaning                                 |
|----------|-----------------------------------------|
| Pending  | Waiting for Admin or Manager review.    |
| Approved | Accepted, and stock has been increased. |
| Rejected | Declined, and stock has not changed.    |

# 9. History

The History page shows a record of inventory activity. It helps the lab understand who changed stock and when the change happened.

## 9.1 Activity Types

| Type    | Meaning                                             |
|---------|-----------------------------------------------------|
| Create  | A new product was added.                            |
| Take    | A user took one or more snacks.                     |
| Restock | Stock increased after an approved restock request.  |
| Adjust  | Stock quantity or alert level was changed manually. |

## 9.2 Filtering History

You can filter History by:

- Activity type.
- Product.
- Date range.
- User, for Admin and Manager users only.

## 9.3 Export History

Select **Export CSV** to download the currently filtered History as a spreadsheet file.

# 10. Analytics

The Analytics page provides simple reports about snack inventory and usage.

| Section       | What It Shows                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------|
| Summary       | Product count, total units, pending restock requests, low-stock alerts, and snacks by category. |
| Restock trend | Restock requests grouped by status.                                                             |
| Take trend    | Weekly snack consumption.                                                                       |
| Top products  | Most frequently taken products.                                                                 |
| Approval time | Average time needed to approve restock requests.                                                |
| Expiry report | Products expiring soon.                                                                         |

Use export buttons, when available, to download report data as a spreadsheet file.

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## 11. Notifications

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Select the bell icon in the header to open the notification panel.

### 11.1 Notification Types

| Type             | Who Receives It   | When It Appears                                             |
|------------------|-------------------|-------------------------------------------------------------|
| Low stock        | Admin and Manager | A product reaches or drops below its low-stock alert level. |
| Restock pending  | Admin and Manager | A user submits a new restock request.                       |
| Restock approved | Requester         | A submitted request is approved.                            |
| Restock rejected | Requester         | A submitted request is rejected.                            |
| Expiry soon      | Admin and Manager | A product is close to its expiry date.                      |

### 11.2 Manage Notifications

- Select a notification to mark it as read.
- Select **Mark all read** to mark all unread notifications as read.
- Unread notifications are highlighted.

Notifications appear inside the application only. Email and SMS notifications are not included in this release.

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## 12. User Management

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User Management is available to Admin users only.

### 12.1 Create a User

1. Open Users.
2. Enter the new user's email, name, password, and role.
3. Select **Create user**.

### 12.2 Edit a User

1. Open Users.
2. Select **Edit** on the user row.
3. Update the user's name, email, role, or password.
4. Save the changes.

### 12.3 Delete a User

1. Open Users.
2. Select **Delete** on the user row.
3. Confirm the action.

You cannot delete your own account while you are signed in.

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## 13. Common Workflows

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### 13.1 Staff Takes a Snack

1. Sign in.
2. Open Inventory.
3. Find the product.
4. Choose the quantity.
5. Select **Take snack**.
6. Check the remaining daily limit if needed.

### 13.2 Request More Stock

1. Sign in.
2. Open Requests.
3. Choose the product.
4. Enter the requested quantity.
5. Submit the request.
6. Check notifications later for the result.

### 13.3 Manager Restocks Inventory

1. Sign in.
2. Open Requests.
3. Find a pending request.
4. Approve the request.
5. Confirm that the stock increased.

### 13.4 Admin Adds Products in Bulk

1. Sign in.
2. Open Inventory.
3. Open the import feature.
4. Download the template.
5. Fill in the product list.
6. Upload the completed file.
7. Review the result.
8. Check Inventory to confirm the imported products.

### 13.5 Investigate Low Stock

1. Check the Dashboard for low-stock alerts.
  2. Open Inventory.
  3. Use the Low Stock filter.
  4. Review the products that need attention.
  5. Create or approve restock requests as needed.
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## 14. Troubleshooting

### 14.1 Cannot Sign In

| Problem                                | What To Do                                                                                |
|----------------------------------------|-------------------------------------------------------------------------------------------|
| The email or password is not accepted. | Check that the information is typed correctly. If it still fails, contact the Admin.      |
| The page keeps returning to Login.     | Sign in again. If the problem continues, clear browser storage or ask the Admin for help. |
| The website does not open.             | Check your network connection and confirm that you are using the correct link.            |

### 14.2 Taking a Snack Fails

| Problem                                  | What It Usually Means                                  |
|------------------------------------------|--------------------------------------------------------|
| The quantity is above the allowed limit. | Staff users must follow the per-take and daily limits. |
| There is not enough stock.               | Choose a smaller quantity or create a restock request. |
| The product cannot be found.             | Refresh the page and search again.                     |

### 14.3 Restock Approval or Rejection Fails

| Problem                                    | What To Do                                                                                     |
|--------------------------------------------|------------------------------------------------------------------------------------------------|
| The request was already handled.           | Refresh the Requests page and check the latest status.                                         |
| The approve or reject button is not shown. | Only pending requests can be approved or rejected. Also confirm that your role has permission. |

### 14.4 Product Import Fails

| Problem                            | What To Do                                             |
|------------------------------------|--------------------------------------------------------|
| Duplicate SKU.                     | Use a different SKU for the new product.               |
| Required information is missing.   | Fill in the required fields in the template.           |
| The unit is not accepted.          | Use one of the supported unit options in the template. |
| Some rows fail but others succeed. | Correct the failed rows and upload them again.         |

### 14.5 Product Image Does Not Show

- Check that the image file or image link is valid.
- Try uploading the image again.
- If the image still does not appear, ask an Admin or Manager to update it.

## 15. Glossary

| Term | Meaning                                                        |
|------|----------------------------------------------------------------|
| SKU  | A unique product code used to identify each snack.             |
| Take | The action of removing snacks from inventory for personal use. |

| Term                  | Meaning                                                                  |
|-----------------------|--------------------------------------------------------------------------|
| Quota                 | The snack limit applied to Staff users.                                  |
| Restock               | Adding more units of a product to inventory.                             |
| Low-stock alert level | The stock number that triggers a low-stock warning.                      |
| History               | A record of inventory actions, such as taking snacks or adjusting stock. |
| User role             | A permission level, such as Admin, Manager, or Staff.                    |
| Pending               | A restock request that is waiting for review.                            |
| Approved              | A restock request that has been accepted.                                |
| Rejected              | A restock request that has been declined.                                |